How to CAD Almost Anything!

Summer 2026 – AeroAstro Workshop

A compressed yet rewarding introduction to the parametric design software PTC Creo, for beginners (no experience at all) and pro-users alike. Come learn how to CAD (computer-aided design) almost anything!

Yes, this could be YOU at the end of the workshop! You'll be equipped with the tools to design cool looking objects such as an iPod classic, an X-wing fighter, a natural trumpet, a Playmobil figure, a LEGO Tower Bridge, a PlayStation 1 controller, a GameBoy, a Pokémon card, and even a Diet Coke can! These are all projects from "How to CAD Almost Anything! – IAP 2026".

Workshop Details

Subject Title:	How to CAD Almost Anything!
Prerequisites:	Willingness to have fun and think outside the box!
Enrollment:	20.
Attendance:	Participants must attend all sessions.
Meeting Rooms:	GIS & Data Lab (first floor of the Rotch Library, 7-238).
Meeting Times:	Wednesdays 3pm – 5pm. There is a total of 8 weekly sessions, starting on 06/03/26 until 07/29/26. The sessions will take place on 06/03, 06/10, 06/17, 06/24, 07/08, 07/15, 07/22, and 07/29.

Instructors: Andy Eskenazi - AeroAstro PhD student (LAE), andyeske@mit.edu.

About: Hailing from the capital of Tango, Steak and Football, Andy is currently a PhD student at MIT AeroAstro trying to make aviation more sustainable. Outside of research, he is passionate about all things mechanical design. At MIT, he has taught the "How to CAD Almost Anything!" series six times, on Solidworks (twice), Fusion 360, Onshape, Siemens NX, Inventor, and now on Creo!

Workshop Description

Ever wondered how musical instruments are designed? How can we generate a computer 3D model of a Trombone, an Ukelele or a Harp? What about designing a Morin Khuur (a traditional Mongolian violin)? Or a vertical piano? In this fun MIT Summer 2026 workshop, you will learn the skills to design all of these, and much more!

Split into 8, 2-hour long sessions, the first half of each session will be spent learning new PTC Creo skills, while the second half will see the application of these new skills through in-class activities. In contrast to traditional mechanical design courses, this workshop places a greater emphasis on the design process itself, understanding how we can plan and best leverage our available tools to arrive to our desired result. Thus, the sessions are less about following the instructions on an engineering drawing, and more about independent thinking and strategizing, reverse engineering an object into a 3D model. This edition of "How to CAD Almost Anything" is themed around musical instruments, which means that all projects (20 total) will involve designing a variety of musical instruments!

Workshop Schedule

#	Month	Day	Date	Outline and Objectives
1	June	W	06/03	**Session 1**: Introduction to PTC Creo. Objective: In this session, we'll get ourselves acquainted with the PTC Creo workspace, and start learning some of the software's most used tools. Session 1's goals include: • Creating sketches (using basic shapes, construction lines, smart-dimensioning, sketch relationships) and understanding planes. • Understanding what it means for a sketch to be fully defined. • Locating and using the different elementary feature commands (boss extrude, boss cut, fillet, chamfer). • Editing sketches and features after creating them. • Coloring parts and changing material properties. Session activity: Using the tools learned on Session 1, we'll design three instruments, namely: • A toy Xylophone with its sticks. • A Pan flute. • A Gong (a percussion instrument).
2	June	W	06/10	**Session 2**: Splines and sketch pictures. Objective: In this session, we'll continue exploring some of the most powerful Creo tools. Session 2's goals include: • Learning how to use the spline tool and project entities. • Learning how to add a picture and sketch on it.

				• Learning how to employ the “master sketch” technique. Session activity: Using the tools learned on Session 2, we'll design three instruments, namely: • An Ukelele. • An electric Guitar. • A Morin Khuur (a Mongolian violin).
3	June	W	06/17	**Session 3**: Advanced extrudes and spines. Objective: In this session, we'll revise the contents learned during Session 1 and Session 2. Session 3's goals include: • Revising some of the previously learned commands, including extrude, fillet, chamfer, sketch picture, and project entities. • Continuing to employ the “master sketch” technique. Session activity: Using the tools learned on Session 3, we'll design three instruments, namely: • A Harmonica. • A traditional Violin. • A Pipa (a Chinese lute).
4	June	W	06/24	**Session 4**: Symmetry, patterns and plane creation. Objective: In this session, we'll focus our attention to symmetry, patterns and planes, and how we can leverage certain tools to simplify the design process. Session 4's goals include: • Understanding how to create a sketch for a revolve. • Learning how to make use of the mirroring and circular patterns tools, both as a sketch and as a feature. • Learning how to create planes, at different angles. Session activity: Using the tools learned on Session 4, we'll design three instruments, namely: • A set of Maracas. • A 4-string Banjo. • A set of wooden Bongos.
5	July	W	07/08	**Session 5**: Loft and sweep. Objective: In this session, we'll explore two very powerful Creo tools, loft and sweep. These tools allow the user to create complicated-looking instruments, like the piping of a horn or the bell of a trombone. Session 5's goals include: • Learning how to use the loft and sweep commands. • Continuing to master previously explored tools, such as revolve, linear/circular patterns and plane creation. Session activity: Using the tools learned on Session 5, we'll design two instruments, namely: • A natural Horn. • A classic Trombone.

6	July	W	07/15	**Session 6**: Advanced loft, sweep and patterns. Objective: In this session, we'll revise the contents learned during Session 4 and Session 5. Session 6's goals include: • Revising some of the previously learned commands, including loft, revolve, sweep, plane creations, patterns, filleting, and material properties. Session activity: Using the tools learned on Session 6, we'll design two instruments, namely: • An Accordion. • A 26-string Harp.
7	July	W	07/22	**Session 7**: Assemblies. Objective: In this session, we'll move towards one of the most powerful features within CAD parametric software, which is that of making assemblies! Session 7's goals include: • Learning how to make an assembly of multiple parts. • Learning how to make an exploded view of an assembly. • Learning how to create an engineering drawing of an assembly (including exploded views). Session activity: Using the tools learned on Session 7, we'll design two instruments, namely: • A Cymbal stand. • A Bell kit with stands.
8	July	W	07/29	**Session 8**: Variable tables and equations. Objective: In this session, we'll investigate two often underappreciated yet extremely useful tools, parameters and equations. These tools allow the user to create various configurations of the same model, which can vary in size, shape, and even color! Session 8's goals include: • Learning how to create parameters and relations. • Creating configurations/family tables of the same model. Session activity: Using the tools learned on Session 8, we'll design two instruments, namely: • A vertical (upright) Piano. • A Euphonium.